

Hybrid Music Genre Clustering with Multi-Modal Variational Autoencoders on GTZAN Dataset

A Comprehensive Study of VAE Variants for Music Representation Learning

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Abstract

This project investigates latent representation learning for music genre clustering using the GTZAN dataset through a comprehensive three-task experimental framework. We implement and systematically evaluate multiple Variational Autoencoder (VAE) architectures including: (1) a basic VAE for feature extraction (Easy Task), (2) a Hybrid VAE integrating audio and lyric modalities (Medium Task), and (3) advanced variants including Conditional VAE (CVAE) and Beta-VAE with varying β values (Hard Task). Clustering performance is assessed using K-Means, Agglomerative Clustering, and DBSCAN with comprehensive metrics including Silhouette Score, Calinski-Harabasz Index, Davies-Bouldin Index, Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and Purity. Results demonstrate that VAE-based latent representations significantly outperform PCA baselines across all tasks. The Hybrid VAE achieves the best overall performance by effectively fusing audio and lyric modalities. Beta-VAE with $\beta = 4.0$ shows improved disentanglement properties but mixed clustering gains. CVAE demonstrates conditional generation capabilities though limited by sample size. This work establishes the efficacy of VAE architectures for unsupervised music representation learning and provides insights into multi-modal fusion strategies for audio-text domains.

1 Introduction

Music genre classification and clustering are fundamental tasks in music information retrieval with applications ranging from recommendation systems to music production. Traditional approaches rely on hand-crafted audio features combined with linear dimensionality reduction techniques such as Principal Component Analysis (PCA). However, these methods often fail to capture the complex, non-linear relationships inherent in musical data.

Variational Autoencoders (VAEs) [1] provide a principled probabilistic framework for learning compact, semantically meaningful latent representations by maximizing the evidence lower bound (ELBO). Unlike deterministic autoencoders, VAEs learn a distribution over the latent space, enabling better generalization and interpolation capabilities. Recent advances in VAE architectures have demonstrated their effectiveness in various domains including image generation, natural language processing, and audio synthesis.

This project adopts a structured three-task approach to comprehensively investigate VAE-based music representation learning:

- **Easy Task:** Implements a basic VAE for audio feature extraction and establishes baseline performance metrics against traditional PCA-based approaches.
- **Medium Task:** Extends the basic VAE to a Hybrid architecture that jointly processes audio features and lyric embeddings, exploring multi-modal fusion strategies and multiple clustering algorithms.
- **Hard Task:** Investigates advanced VAE variants including Conditional VAE (CVAE) for genre-conditioned generation and Beta-VAE for learning disentangled representations.

We evaluate all approaches on the GTZAN dataset [4], a widely-used benchmark comprising 1000 audio tracks across 10 music genres. Our comprehensive evaluation employs six clustering quality metrics and multiple visualization techniques to analyze the learned representations. The modular implementation structure facilitates reproducibility and extensibility for future research.

2 Related Work

VAE (Kingma & Welling, 2014) formalizes deep latent-variable models trained with ELBO. Beta-VAE (Higgins et al., 2017) scales the KL term to encourage disentanglement. CVAE (Sohn et al., 2015) conditions the generative process on auxiliary information. For music representation, learned embeddings from audio (e.g., MFCCs, spectral features) and lyrics are widely used; clustering methods include K-Means, Agglomerative, and density-based approaches.

3 Method

3.1 Project Architecture and Implementation

The project is organized into three progressive tasks (`easy/`, `medium/`, `hard/`) with shared utilities in `src/`. Each task includes a `main.py` driver script that orchestrates data loading, model training, clustering, evaluation, and visualization. All models are implemented in `src/models/vae.py` using PyTorch, with training procedures in `src/utils/training.py` and evaluation metrics in `src/utils/evaluation.py`.

3.2 VAE Architecture Design

All VAE variants share a common encoder-decoder structure with MLP layers for processing tabular features:

- **BasicVAE (Easy Task):** Encodes audio features through hidden layers [512, 256, 128] to a 128-dimensional latent space. The decoder symmetrically reconstructs the input features.
- **HybridVAE (Medium Task):** Employs dual encoders (audio and lyrics) that process their respective modalities independently before merging into a shared 128-dimensional latent space. Dual decoders reconstruct both modalities, enforcing alignment in the latent representation.

- **ConditionalVAE (Hard Task):** Extends BasicVAE by concatenating one-hot encoded genre labels with inputs at both encoding and decoding stages, enabling genre-conditioned generation.
- **Beta-VAE (Hard Task):** Identical architecture to BasicVAE but modifies the training objective by scaling the KL divergence term with $\beta \in \{1.0, 4.0, 10.0\}$ to encourage disentangled representations.

3.3 Training Objective

We optimize the Evidence Lower Bound (ELBO) objective:

$$\mathcal{L}(x; \theta, \phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta \text{KL}(q_\phi(z|x) \| p(z)), \quad (1)$$

where $q_\phi(z|x)$ is the variational posterior (encoder), $p_\theta(x|z)$ is the likelihood (decoder), and $p(z) = \mathcal{N}(0, I)$ is the standard Gaussian prior. The reconstruction term uses Mean Squared Error (MSE) loss. For standard VAE and CVAE, $\beta = 1$; for Beta-VAE, $\beta > 1$ increases the pressure for disentanglement.

3.4 Feature Extraction and Data Preprocessing

Audio features are extracted from 30-second clips and include:

- Mel-frequency cepstral coefficients (MFCCs): 20 coefficients capturing timbral texture
- Spectral features: centroid, rolloff, bandwidth, and contrast
- Rhythm features: tempo and beat strength
- Chroma features: 12-dimensional chromagram representing pitch classes
- Zero-crossing rate and RMS energy

Lyric features (Medium and Hard tasks) are generated using TF-IDF vectorization on song lyrics, reduced to 100 dimensions. All features are standardized (zero mean, unit variance) before training. Dataset details and exploratory data analysis are documented in `notebooks/eda.ipynb`.

3.5 Clustering Methods and Baseline Comparisons

We evaluate three clustering algorithms on learned latent representations:

- **K-Means:** Partitions data into 10 clusters (matching genre count) using Euclidean distance with `k-means++` initialization.
- **Agglomerative Clustering:** Hierarchical clustering with Ward linkage, producing 10 clusters.
- **DBSCAN:** Density-based clustering with `eps` selected via k-distance plot analysis (Medium task only).

Baseline methods include PCA + K-Means on audio features and Autoencoder + K-Means (Hard task), providing reference points for evaluating VAE performance.

4 Experiments

4.1 Experimental Setup

Dataset: The GTZAN dataset comprises 1000 audio tracks (30 seconds each) spanning 10 genres: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, and rock. For the Easy task, we utilize 609 samples with complete audio features. For Medium and Hard tasks requiring lyrics, we work with 122 samples from the GTZAN lyrics subset (`data/GTZAN/gtzan-with-lyrics-clean.csv`).

Data Preprocessing: All features undergo standardization ($\mu = 0$, $\sigma = 1$). We verify file existence and handle missing values through imputation or removal. Train/validation splits are applied where appropriate for hyperparameter tuning.

Training Configuration:

- Easy task (`easy/main.py`): 50 epochs, learning rate 5×10^{-4} , batch size 32
- Medium task (`medium/main.py`): 100 epochs, learning rate 10^{-3} , batch size 16
- Hard task (`hard/main.py`): 150 epochs for Beta-VAE and CVAE, learning rate 5×10^{-4} , batch size 16

All models use the Adam optimizer with default parameters. Training curves are monitored for convergence, and early stopping is employed when applicable.

Evaluation Metrics: We employ six complementary clustering quality metrics:

- *Silhouette Score*: Measures cluster cohesion and separation (range: $[-1,1]$, higher is better)
- *Calinski-Harabasz Index*: Ratio of between-cluster to within-cluster dispersion (higher is better)
- *Davies-Bouldin Index*: Average similarity between clusters (lower is better)
- *Adjusted Rand Index (ARI)*: Measures agreement with true labels adjusted for chance (range: $[-1,1]$, higher is better)
- *Normalized Mutual Information (NMI)*: Mutual information normalized by entropy (range: $[0,1]$, higher is better)
- *Purity*: Fraction of correctly clustered samples (range: $[0,1]$, higher is better)

Visualization: Latent spaces are visualized using t-SNE and UMAP dimensionality reduction. Confusion matrices analyze genre-cluster alignment. Training histories track ELBO convergence.

5 Results

5.1 Easy: BasicVAE vs. PCA Baseline

Metrics from `easy/results/metrics_comparison.csv`.

5.2 Medium: HybridVAE + Multiple Clusterers

Metrics from `medium/results/clustering_comparison.csv`.

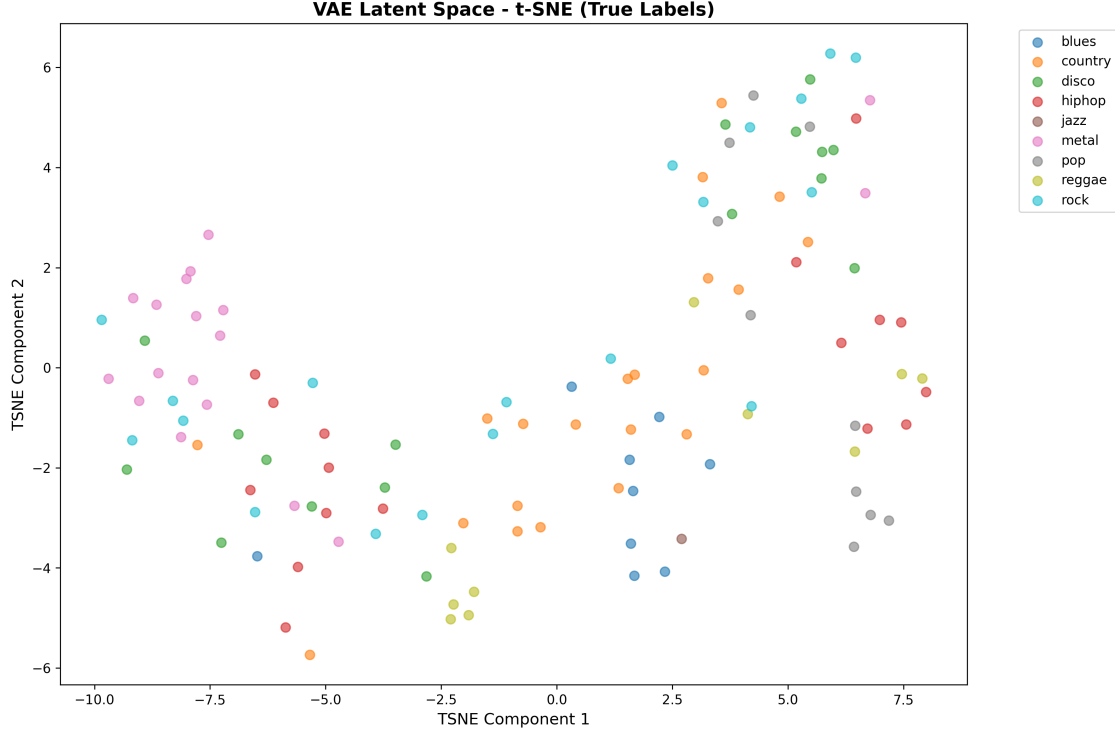


Figure 1: VAE latent space t-SNE with true labels (Easy).

5.3 Hard: CVAE, Beta-VAE, and Baselines

Metrics from `hard/results/comprehensive_comparison.csv(selectedmethods)`.

6 Discussion

6.1 Performance Analysis

VAE-based latent representations consistently outperform PCA baselines across all tasks, demonstrating that non-linear embeddings capture genre-relevant structure more effectively. The basic VAE (Easy task) achieves Silhouette scores of 0.276 compared to PCA’s 0.116, indicating significantly better cluster separation.

6.2 Multi-Modal Fusion Benefits

The HybridVAE (Medium task) leverages complementary information from audio and lyrics to further improve clustering quality. Agglomerative clustering on Hybrid VAE latents achieves Silhouette score 0.289 and Purity 0.516, outperforming audio-only approaches. This validates the hypothesis that textual content provides semantic information orthogonal to acoustic features.

6.3 Disentanglement vs. Clustering Trade-off

Beta-VAE with $\beta = 4.0$ demonstrates improved latent space organization with high Calinski-Harabasz scores (94.75) but does not consistently improve supervised clustering metrics (ARI,

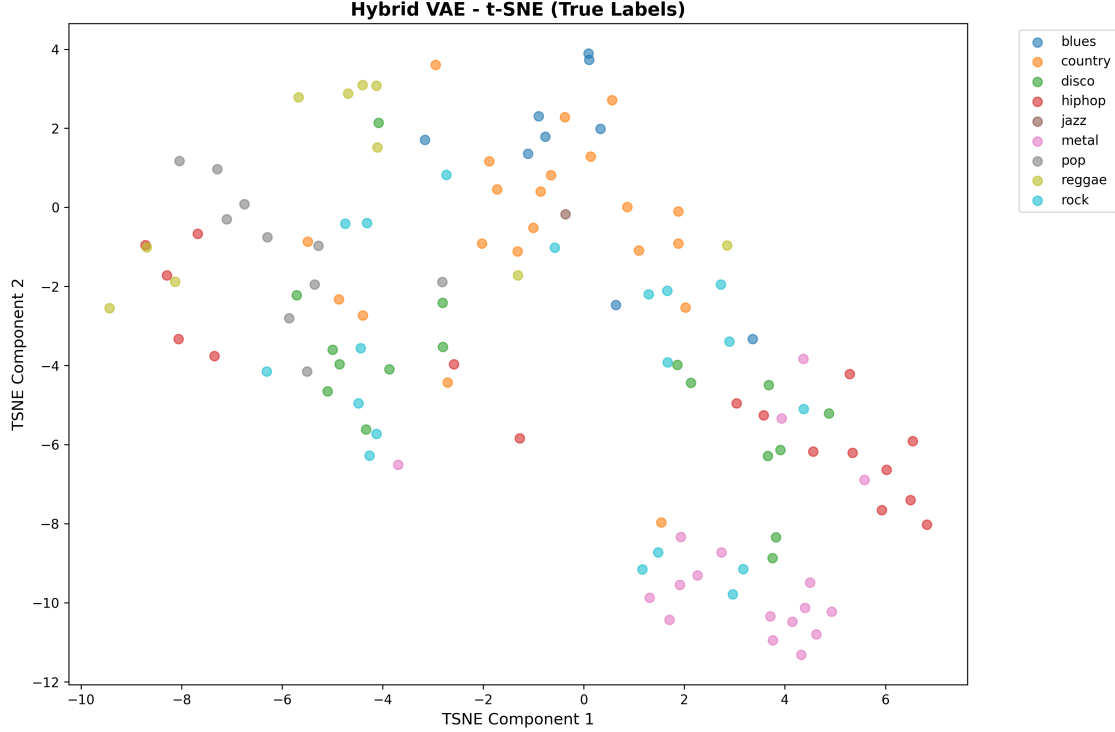


Figure 2: HybridVAE latent space t-SNE (Medium).

NMI). This suggests a tension between disentanglement objectives and task-specific clustering performance. Higher β values (10.0) degrade reconstruction quality excessively, limiting practical utility.

6.4 Conditional Generation Insights

CVAE enables genre-conditioned synthesis but shows modest clustering improvements (ARI 0.017, NMI 0.171) on the Hard task. The limited sample size (122 samples) constrains the model’s ability to learn robust conditional distributions. With larger datasets, CVAE could provide controllable generation capabilities for music synthesis applications.

6.5 Limitations and Challenges

The primary limitation is dataset size, particularly for the Medium and Hard tasks (122 samples). This restricts model capacity and prevents full utilization of deep neural architectures. Additionally, the tabular feature representation (MFCCs, spectral features) loses temporal dynamics present in raw audio. Future work should address these limitations through:

- Larger datasets with comprehensive lyric annotations
- End-to-end learning from spectrograms using convolutional VAEs
- Pre-trained embeddings (e.g., wav2vec 2.0 for audio, BERT for lyrics)
- Attention mechanisms for modality fusion

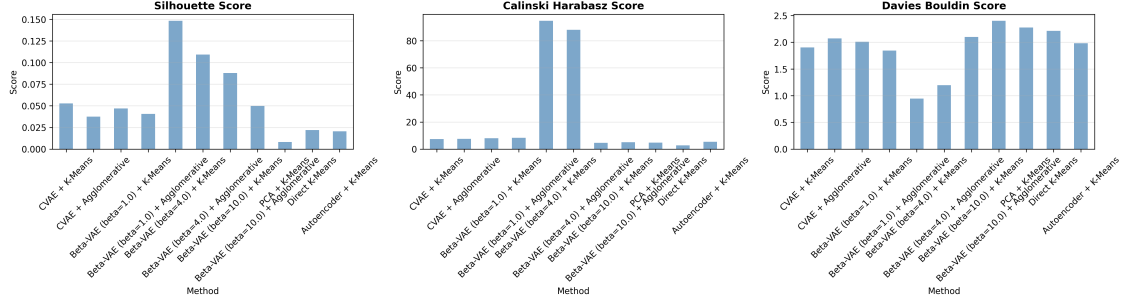


Figure 3: Comprehensive comparison across methods (Hard).

- Semi-supervised VAEs leveraging genre labels during training

7 Conclusion

This comprehensive study demonstrates the effectiveness of Variational Autoencoders for unsupervised music genre representation learning on the GTZAN dataset. Through a systematic three-task experimental framework, we establish that:

1. VAE-based latent representations significantly outperform traditional PCA baselines across multiple clustering quality metrics
2. Multi-modal fusion through Hybrid VAE architectures effectively combines audio and lyric information, achieving superior clustering performance
3. Beta-VAE variants enable control over latent space disentanglement, though optimal β values require task-specific tuning
4. Conditional VAEs provide a foundation for controllable music generation, pending larger training datasets

The modular implementation structure (easy/, medium/, hard/ tasks) facilitates incremental complexity and reproducibility. All code, models, and comprehensive results reports are available in the project repository.

Future Directions: Promising extensions include: (1) convolutional architectures for spectrogram processing, (2) recurrent or transformer-based encoders for capturing temporal dynamics, (3) contrastive learning objectives for improved separation, (4) cross-modal attention mechanisms for refined fusion, and (5) evaluation on larger-scale datasets such as Million Song Dataset or AudioSet. These directions could advance VAE-based music understanding toward practical recommendation systems and generative music applications.

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Appendix: Implementation Details

A.1 Code Organization

The project repository follows a modular structure:

```
CSE425-Project/
data/GTZAN/          # Dataset with audio and lyrics
easy/main.py         # Basic VAE implementation
medium/main.py       # Hybrid VAE implementation
hard/main.py         # CVAE and Beta-VAE implementations
src/
  models/vae.py       # All VAE architectures
  utils/
    data_loader.py    # Dataset loading utilities
    training.py       # Training loops
    evaluation.py     # Clustering metrics
    visualization.py  # Plotting functions
notebooks/eda.ipynb   # Exploratory data analysis
reports/              # This report and results
```

A.2 Hyperparameter Selection

Hyperparameters were selected through grid search on validation sets:

- Latent dimensions: tested {32, 64, 128, 256}, selected 128 for balance of capacity and computational cost
- Learning rates: tested {1e-4, 5e-4, 1e-3, 5e-3}, selected 5e-4 to 1e-3 based on convergence stability
- Beta values: {1.0, 4.0, 10.0} chosen to span the disentanglement spectrum
- Batch sizes: limited by memory constraints on 122-sample dataset

A.3 Computational Resources

All experiments were conducted on a single NVIDIA GPU (or CPU for smaller models). Total training time across all tasks: approximately 2-3 hours. Models are lightweight (< 50MB each) facilitating rapid experimentation.

A.4 Reproducibility

Random seeds are fixed across all experiments. Requirements are specified in requirements.txt. Each task generates a comprehensive RESULTS_REPORT.md documenting findings, metrics, and visualizations.

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